Results in Engineering 25 (2025) 103612

Contents lists available at ScienceDirect

Results in Engineering

journal homepage: www.sciencedirect.com/journal/results-in-engineering

Static stability of functionally graded porous nanoplates under uniform and non-uniform in-plane loads and various boundary conditions based on the nonlocal strain gradient theory

Ihab Omar [a], Thamer Marhoon [b], Shahram Babadoust [c], Akram Shakir Najm [d], Mostafa Pirmoradian [e,*], Soheil Salahshour [f,g,h], S. Mohammad Sajadi [i]

[a] Air Conditioning Engineering Department, Faculty of Engineering, Warith Al-Anbiyaa University, Karbala 56001, Iraq
[b] Department of Chemical Engineering and Petroleum Refining, Kut University College, Kut, Wasit 52001, Iraq
[c] Department of Medical Biochemical Analysis, Cihan University-Erbil, Erbil, Kurdistan Region, Iraq
[d] University of Technology- Iraq
[e] Department of Mechanical Engineering, Khomeinishahr branch, Islamic Azad University, Khomeinishahr, Iran
[f] Faculty of Engineering and Natural Sciences, Istanbul Okan University, Istanbul, Turkey
[g] Faculty of Engineering and Natural Sciences, Bahcesehir University, Istanbul, Turkey
[h] Faculty of Science and Letters, Piri Reis University, Tuzla, Istanbul, Turkey
[i] Department of Chemistry, Payam e Noor University, Saqqez Branch, Saqqez, Kurdistan, Iran

A R T I C L E I N F O

Keywords:
Functionally graded porous nanoplates
Elastic buckling
Nonlocal strain gradient theory
Four-variable refined plate theory

A B S T R A C T

This work examines the buckling behavior of functionally graded porous nanoplates embedded in elastic media. Size effects are added to the nanoplate constitutive equations using nonlocal strain gradient theory. The four-variable refined plate theory is employed for nanoplate modeling. This theory assures stress-free conditions on both sides of the nanoplate and has less uncertainty than high-order shear deformation theories. It is postulated that the nanoplate experiences in-plane compressive loads, which may have both linear and nonlinear distributions. Additionally, uniform and non-uniform porosity distributions are considered. The governing partial differential equations are extracted using the notion of the minimal total potential energy. Following this, the Galerkin method is employed to solve these equations utilizing trigonometric shape functions. Simple, clamped, and combined boundary conditions for nanoplate edges are studied. Once the governing algebraic equations were extracted, the critical buckling load of the nanoplate is determined. To conduct a validation study, the obtained data are juxtaposed with the findings of previous studies, revealing a notable level of concurrence. After the critical buckling load has been ascertained, an inquiry is undertaken to assess the influence of various parameters including nonlocal and length scale parameters, boundary conditions, porosity distribution type, in-plane loading type, geometric dimensions of the nanoplate, and stiffness of the elastic environment, on the static stability of nanoplates.

1. Introduction

Nanostructures have garnered significant interest from researchers worldwide in recent years because of their extensive use in nanoelectromechanical systems [1,2]. An example of a two-dimensional nanostructure is the nanoplate, which is defined by its nanoscale thickness and larger lateral dimensions. Because of their large surface area, improved reactivity, and adjustable optical features, these materials find use in a wide range of applications, including electronics, catalysis, and biomedical domains. Various synthesis processes yield nanoplates with varying sizes and forms, allowing for fine-tuning for specific applications. Nanoplates can be made of a wide variety of materials. One example is functionally graded porous (FGP) nanoplates, which show different levels of porosity and composition as the thickness increases. A nanoplate experiences buckling when it is subjected to compressive loads and subsequently deforms or fails abruptly. Due to the distinctive composition and porosity distribution of FGP nanoplates [3–6], their behavior under buckling conditions is of considerable

* Corresponding author.
E-mail address: m.pirmoradian@iaukhsh.ac.ir (M. Pirmoradian).

https://doi.org/10.1016/j.rineng.2024.103612
Received 28 July 2024; Received in revised form 26 November 2024; Accepted 2 December 2024
Available online 5 December 2024

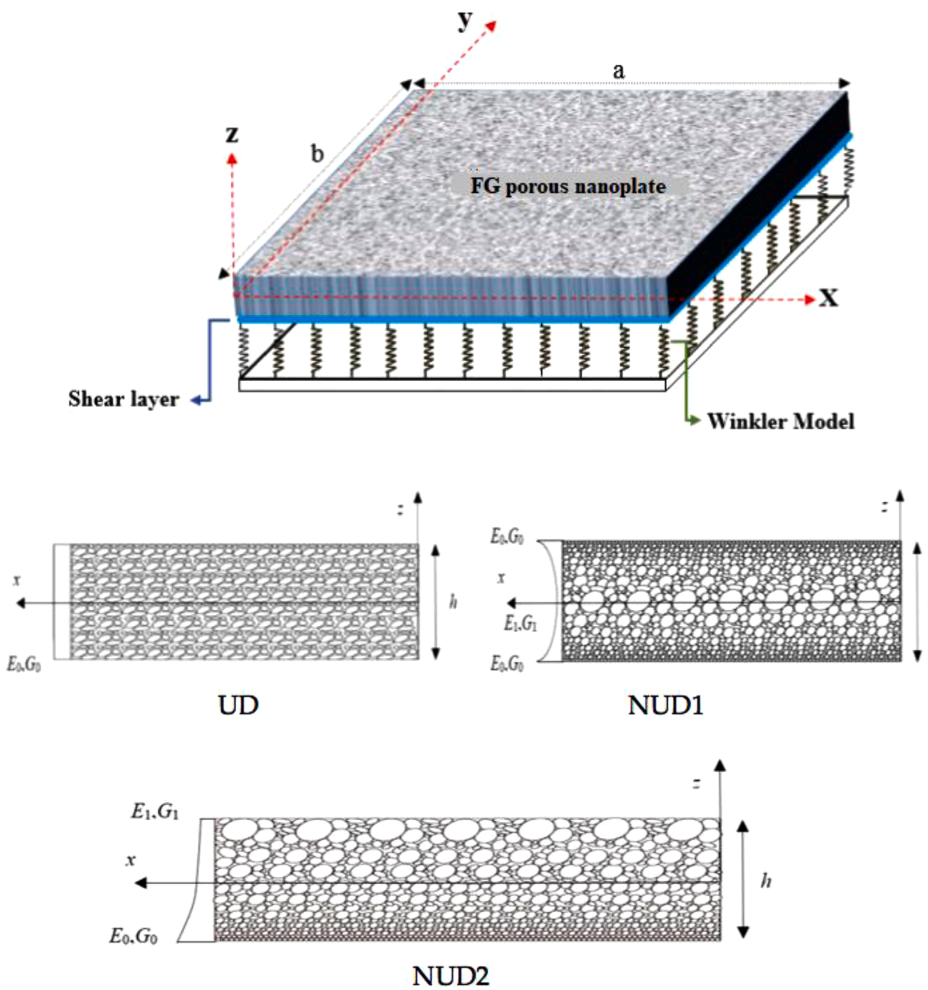

Fig. 1. Schematic of the FGP nanoplate and the desired type of porosities.

interest. The mode and critical buckling load (CBL) of a nanoplate can be substantially impacted by the variation in material properties and pore distribution that occurs throughout the nanoplate.

For investigating materials at the atomic and molecular level, experiments conducted at these dimensions provide the highest degree of certainty. The primary challenges associated with this approach pertain to the intricate task of regulating the experimental parameters at such a scale, alongside the substantial financial burdens and the labor-intensive nature of the methodology. Atomic simulation presents itself as an alternative approach for investigating structures at a microscopic level. This approach examines the behavior of atoms and molecules by taking into account the intermolecular and interatomic influences on their motion. Utilizing this approach, particularly in cases where the problem exhibits significant deformation or involves a scale beyond several atoms, incurs substantial computational expenses and lacks economic viability. In light of the aforementioned constraints, scholars have sought more straightforward approaches in the study of nanostructures. Consequently, the utilization of continuum mechanics has emerged as a potential option for modeling nanostructures. On the nanoscale, the mechanical properties are influenced by the structure's dimensions. Consequently, it is necessary to employ size-dependent elasticity field theories for modeling this issue. In nonlocal elasticity theories, the use of size-dependent effects entails manipulating the governing structural equations. Researchers have proposed several size-dependent continuum theories. A few examples of these theories are the nonlocal theory [7], the modified couple stress theory [8], the strain gradient elasticity theory [9], and the nonlocal strain gradient theory (NSGT) [10]. The NSGT is a contemporary theoretical framework that incorporates the nonlocal stress field and strain gradient effects at the same time. This is achieved through the utilization of two scale factors. Consequently, the outcomes of this theory exhibit a strong concurrence with both experimental findings and molecular dynamics simulations.

The buckling features of FGP nanoplates play an essential role in a widespread range of sophisticated applications spanning diverse industries. FGP nanoplates are crucial for various important applications, such as aerospace engineering, biomedical devices, micro-electromechanical systems (MEMS), automotive engineering, energy storage and conversion, and civil infrastructure [11]. Looking at how FGP nanoplates behave when they buckle in different situations gives engineers and researchers the chance to make designs better, make structures more reliable, and speed up technological progress in these important fields. In this regard, Khaniki et al. [12] examined the buckling characteristics of nonuniform small-scale beams by applying the NSGT. The study shows that the presence of nonuniformity can greatly affect CBLs. Dang and Nguyen [13] used the NSGT to solve the nonlinear vibration and buckling issues of the FGP micro-beam. They explored both even and uneven porosity distribution models. The FGP micro-beam's buckling and nonlinear free vibration responses were examined and discussed with regard to the length-thickness ratio, porosity distribution factor, nonlocal parameter (nl_parameter), power-law index, material length scale (LS) parameter, and elastic foundation coefficients. In addition, Khorshidi and Fallah [14] studied the buckling of FGP nanoplate. They investigated small-scale effects using the nonlocal elasticity theory. Plate material characteristics were supposed to vary in power-law form in thickness. The influence of power law indexes, nl_parameters, and aspect ratio on rectangular FG nanoplate CBL was discussed. Alghanmi [15] examined static analysis of FGP nanoplates. The nanoplate was modeled using NSGT and the four-variable theory. Two different types of porosity distribution were considered. The effects of nonlocal and strain gradient factors, aspect ratio, and porosity were examined. Zenkour and Aljadani [16] studied the buckling of FGP plates. They presented three models of FGP plates: isotropic, homogeneous core, and homogenous skins. Based on the modified polynomial law, FGP material attributes varied with layer thickness. Thickness stretching, mechanical loadings, porosity coefficients (Poros_Coef), gradient indices, geometric characteristics, and layer thickness ratios were examined.

Wang and Zhang [16] investigated FGP metal foam nanoplate buckling using both nonlocal elasticity theory and the refined plate theory. Golmakani and Vahabi [17] used the nonlocal elasticity theory to investigate the buckling of FG annular graphene sheets. They investigated the impact of small-scale phenomena, geometrical parameters, grading index, buckling mode, and boundary conditions (BCs) on the CBL. Barati and Shahverdi [18] utilized NSGT to conduct forced vibration evaluation of FGP nanoplates subjected to uniform loads. The study revealed that various factors, including porosities, elastic foundation, nl_parameter, excitation frequency, material gradation, and strain gradient parameter, exert a substantial influence on the forced vibration characteristics of the nanoplate. Mohseni and Naderi [19] used higher-order shear and normal deformable plate theory (HOSNDT) to analyze the stability of thick rectangular FGP plates under in-plane loadings. It was supposed that plate material and porosity varied with thickness. Compared to the HSDT, the HOSNDT reduced plate CBL, and its effect increased with plate thickness. Furthermore, HOSNDT had a stronger influence on plates under biaxial compression loads than uniaxial loads. With smaller power-law index parameters, the porosity effect increased. Nonlocal heat conduction was used to examine the buckling of FG Euler-Bernoulli beams at non-uniform temperatures by Xu et al. [20]. The expected numerical results show that the nl_parameter enhancement improves the dimensionless CBLs in all BCs. Malikan et al. [21] studied rectangular FGP nanoplate buckling. Large plate deformations and thickness changes after deflection were discussed. Even and uneven distributions for porosity were assumed. By using nonlocal elasticity theory, nonlocal equations were derived. The plate was softer with even porosity and closer to the perfect material with uneven porosity.

In the examination of FG nanoplates, many plate theories can be employed to investigate their mechanical characteristics. These theories encompass classical plate theory [22–24], first-order shear deformation plate theory [25–28], higher-order plate theories [29–31], and the four-variable refined theory [32–35]. By incorporating both shear deformation and bending effects into a balanced framework, the four-variable refined theory provides a rigorous and adaptable method for analyzing the behavior of plate structures. By integrating supplementary variables, this theory effectively captures shear deformation, rendering it ideally suited for the analysis of plates exhibiting diverse thicknesses, material characteristics, and loading circumstances [36–38]. Therefore, while this theory requires more computational resources, it provides enhanced precision, particularly when examining FG nanoplates characterized by intricate material distributions. According to a review of the literature on porous structures, the authors have not come across any research examining the buckling of FGP rectangular nanoplates under uniform and non-uniform in-plane loadings using the four-variable refined theory of plates on the side NSGT. Furthermore, the phenomenon of buckling in porous nanoplates holds significant importance in the realm of nanoscale sensor design. Hence, this study aims to examine the various facets of the buckling issue of FGP nanoplates exposed to both linear and non-linear in-plane loads.

2. Problem formulation

This section presents the derivation of the equations that govern the mechanical behavior of an FGP nanoplate situated on an elastic media, under both uniform and non-uniform in-plane loadings. This study employs the four-variable refined plate theory to model the nanoplate and incorporates the size effect into the governing equations using the NSGT. This study employs the notion of the minimum total potential energy to develop the governing equations. The nanostructure depicted in Fig. 1 consists of an FGP rectangular nanoplate with length a, width b, and thickness h. Three distinct porosity distributions are examined, as depicted in Fig. 1: uniform distribution (UD), first type non-uniform porosity distribution (NUD1), and second type non-uniform porosity distribution (NUD2). Similar to functionally graded materials, NUDs of the first and second categories exhibit graded mechanical properties. The elastic modulus of porous materials varies with thickness, based on the usual mechanical characteristics of open-cell metal foam.

Young's modulus E, shear modulus G, and density do not change across the nanoplate's thickness in the scenario of UD. The NUD1 exhibits maximum values for the aforementioned properties at the top and bottom surfaces, whereas these values are lowest at the center surface of the nanoplate. For NUD2, the elastic modulus, shear modulus, and density exhibit a progressive transition from the upper level to the lower level. Consequently, the bottom surface of the nanoplate possesses the highest values, while the upper surface possesses the lowest values. The following calculation is performed for all three types of porosity under consideration [39–41]:

- UD:

$$E(z) = E_0(1-\lambda\eta), \tag{1}$$

$$G(z) = G_0(1-\lambda\eta), \tag{2}$$

$$\rho(z) = \rho_0\sqrt{(1-\lambda\eta)}, \tag{3}$$

- NUD1:

$$E(z) = E_0\left(1-\lambda\cos\left(\frac{\pi z}{h}\right)\right), \tag{4}$$

$$G(z) = G_0\left(1-\lambda\cos\left(\frac{\pi z}{h}\right)\right), \tag{5}$$

$$\rho(z) = \rho_0\left(1-\zeta\cos\left(\frac{\pi z}{h}\right)\right), \tag{6}$$

- NUD2:

$$E(z) = E_0\left(1-\lambda\cos\left(\frac{\pi z}{2h}+\frac{\pi}{4}\right)\right), \tag{7}$$

$$G(z) = G_0\left(1-\lambda\cos\left(\frac{\pi z}{2h}+\frac{\pi}{4}\right)\right), \tag{8}$$

$$\rho(z) = \rho_0\left(1-\zeta\cos\left(\frac{\pi z}{2h}+\frac{\pi}{4}\right)\right), \tag{9}$$

where E_0 and G_0 stand for the maximum values of Young's and shear moduli, respectively. In addition, λ expresses the Poros_Coef. The elastic and shear moduli of an open-cell porous material concerning the porosity parameter may be determined as follows:

$$\lambda = 1-\frac{E_1}{E_0} = 1-\frac{G_1}{G_0}, \tag{10}$$

where E_1 and G_1 are the least values of Young's and shear moduli,

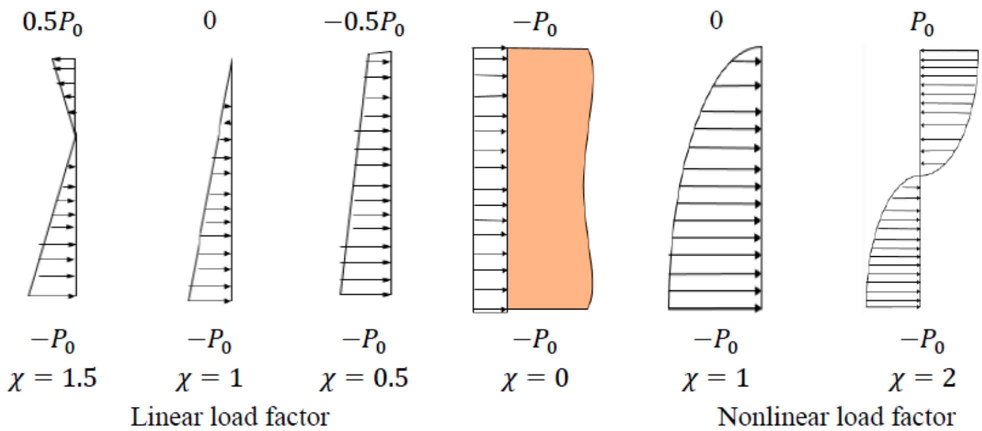

Fig. 2. Uniform and non-uniform in-plane loads.

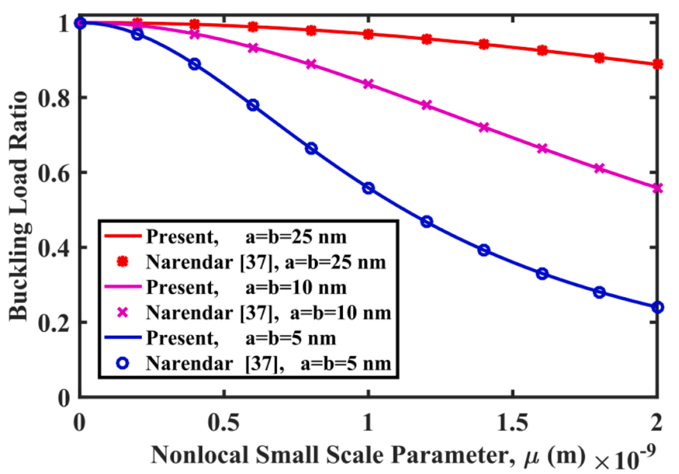

Fig. 3. Variations of CBL ratio with nl_parameter for different dimensions of graphene nanoplate.

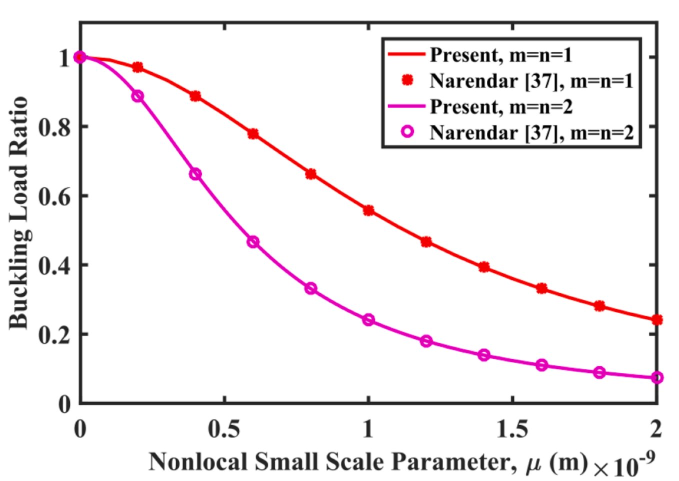

Fig. 4. CBL ratio changes with nl_parameter for different buckling mode numbers.

respectively. In addition, in the above equations, coefficients η and ζ are functions of λ which are defined according to Eqs. (11) and (12) [39]:

$$\eta = \frac{1}{\lambda} - \frac{1}{\lambda}\left(\frac{2}{\pi}\sqrt{1-\lambda} - \frac{2}{\pi} + 1\right)^2, \tag{11}$$

$$\zeta = 1 - \sqrt{1-\lambda} \tag{12}$$

According to the four-variable refined plate theory [42], the displacement field is:

$$\begin{aligned} u_1(x,y,z) &= u(x,y) - z\frac{\partial w_b(x,y)}{\partial x} - f(z)\frac{\partial w_s(x,y)}{\partial x}, \\ u_2(x,y,z) &= v(x,y) - z\frac{\partial w_b(x,y)}{\partial y} - f(z)\frac{\partial w_s(x,y)}{\partial y}, \\ u_3(x,y,z) &= w_b(x,y) + w_s(x,y), \end{aligned} \tag{13}$$

where u_1, u_2, and u_3 are the displacement of an arbitrary point of the nanoplate in the directions of x, y, and z, respectively. Moreover, u and v stand for the displacements of the middle surface of the nanoplate in the directions of x and y, while w_b and w_s are bending and shear deformations, respectively. Additionally, $f(z)$ is a shape function that expresses the distribution of shear stresses and transverse shear strains along the thickness of the nanoplate, and it is chosen in such a way as to provide stress-free BCs on the nanoplate's upper and lower surfaces. In this study, this function is considered with sinusoidal changes as $f(z) = z - h/\pi \sin(\pi z/h)$.

Small strains are generated when deformations are small, as indicated by Eq. (14) [43]:

$$\begin{gathered} \varepsilon_{xx} = \frac{\partial u_1}{\partial x}, \quad \varepsilon_{yy} = \frac{\partial u_2}{\partial y}, \quad \varepsilon_{zz} = \frac{\partial u_3}{\partial z}, \\ \gamma_{xy} = \frac{1}{2}\left(\frac{\partial u_1}{\partial y} + \frac{\partial u_2}{\partial x}\right), \quad \gamma_{xz} = \frac{1}{2}\left(\frac{\partial u_1}{\partial z} + \frac{\partial u_3}{\partial x}\right), \quad \gamma_{yz} = \frac{1}{2}\left(\frac{\partial u_2}{\partial z} + \frac{\partial u_3}{\partial y}\right), \end{gathered} \tag{14}$$

where ε_{xx}, ε_{yy}, and ε_{zz} represent normal strains and γ_{xy}, γ_{xz} and γ_{yz} stand for shear strains.

Substituting the displacement field equations into Eq. (14) leads to strains for small deformations as:

$$\begin{gathered} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{Bmatrix} = \begin{Bmatrix} \varepsilon_{xx}^{(0)} \\ \varepsilon_{yy}^{(0)} \\ \gamma_{xy}^{(0)} \end{Bmatrix} + z\begin{Bmatrix} \kappa_{xx}^{b} \\ \kappa_{yy}^{b} \\ \kappa_{xy}^{b} \end{Bmatrix} + f(z)\begin{Bmatrix} \kappa_{xx}^{s} \\ \kappa_{yy}^{s} \\ \kappa_{xy}^{s} \end{Bmatrix}, \\ \begin{Bmatrix} \gamma_{yz} \\ \gamma_{xz} \end{Bmatrix} = \begin{Bmatrix} \gamma_{yz}^{s} \\ \gamma_{xz}^{s} \end{Bmatrix}, \end{gathered} \tag{15}$$

where

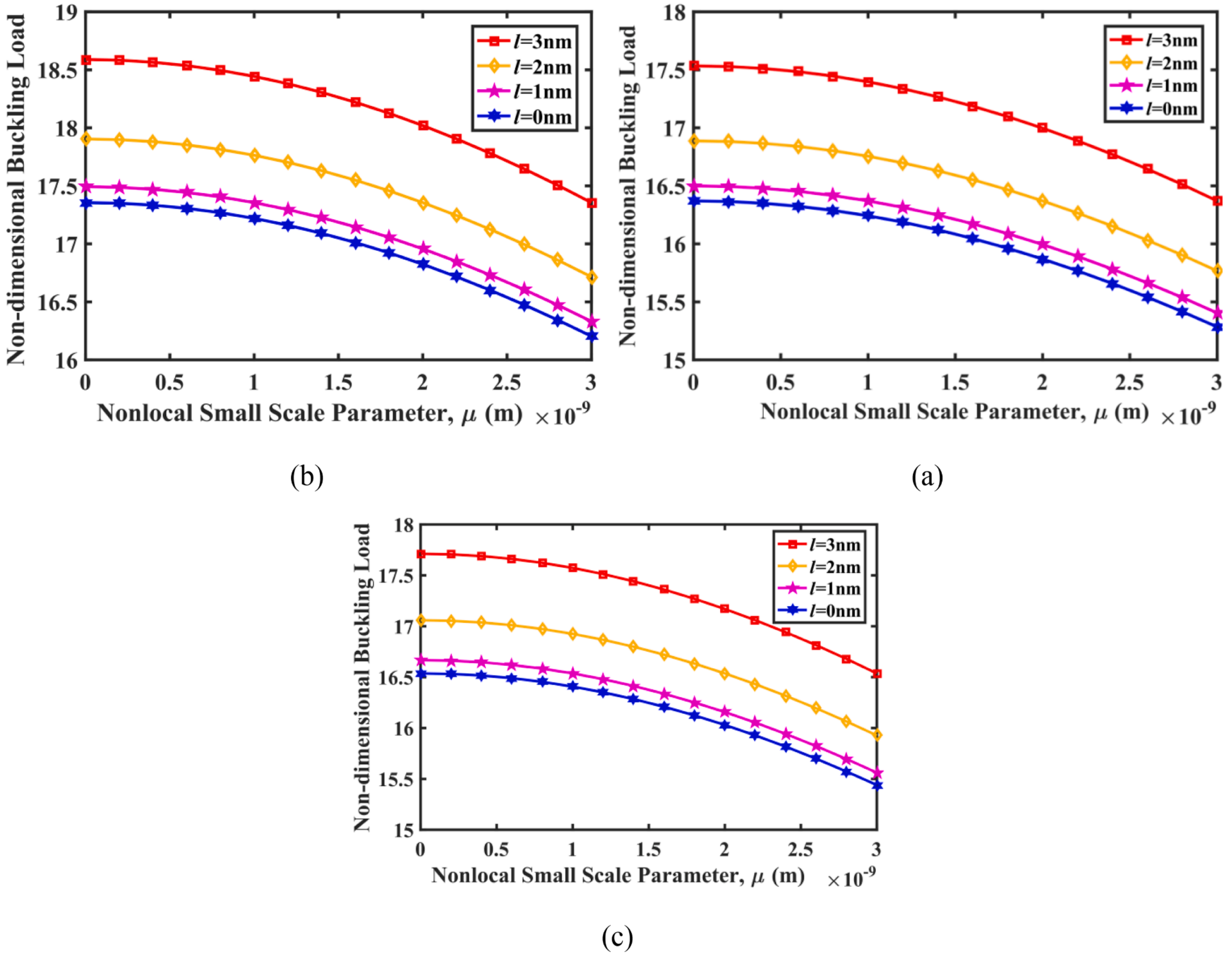

Fig. 5. Dimensionless CBL changes versus nl_parameter for different LS parameters; (a) UD, (b) NUD1 and (c) NUD2.

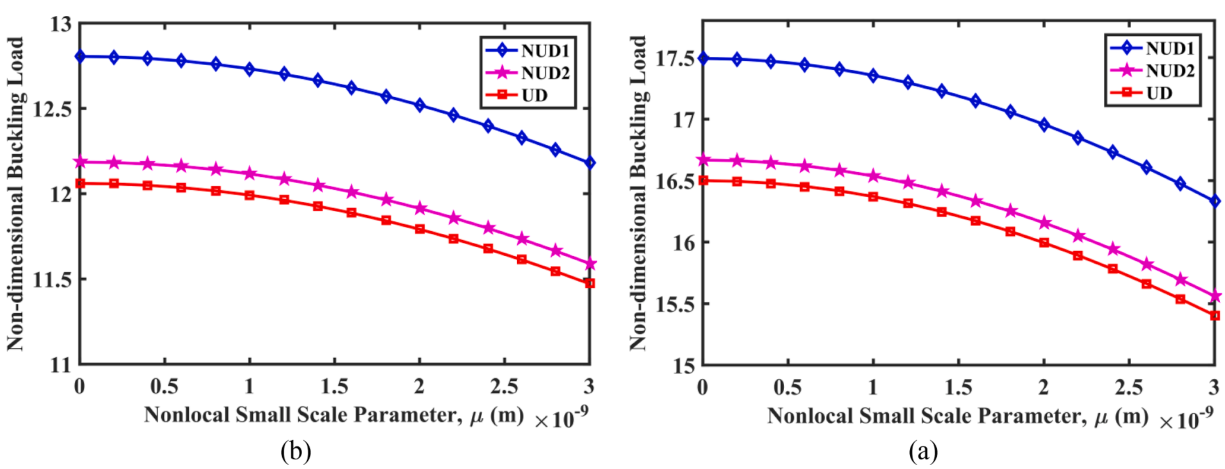

Fig. 6. Dimensionless CBL changes versus nl_parameter for different porosity distributions; (a) Square nanoplate and (b) Rectangular nanoplate.

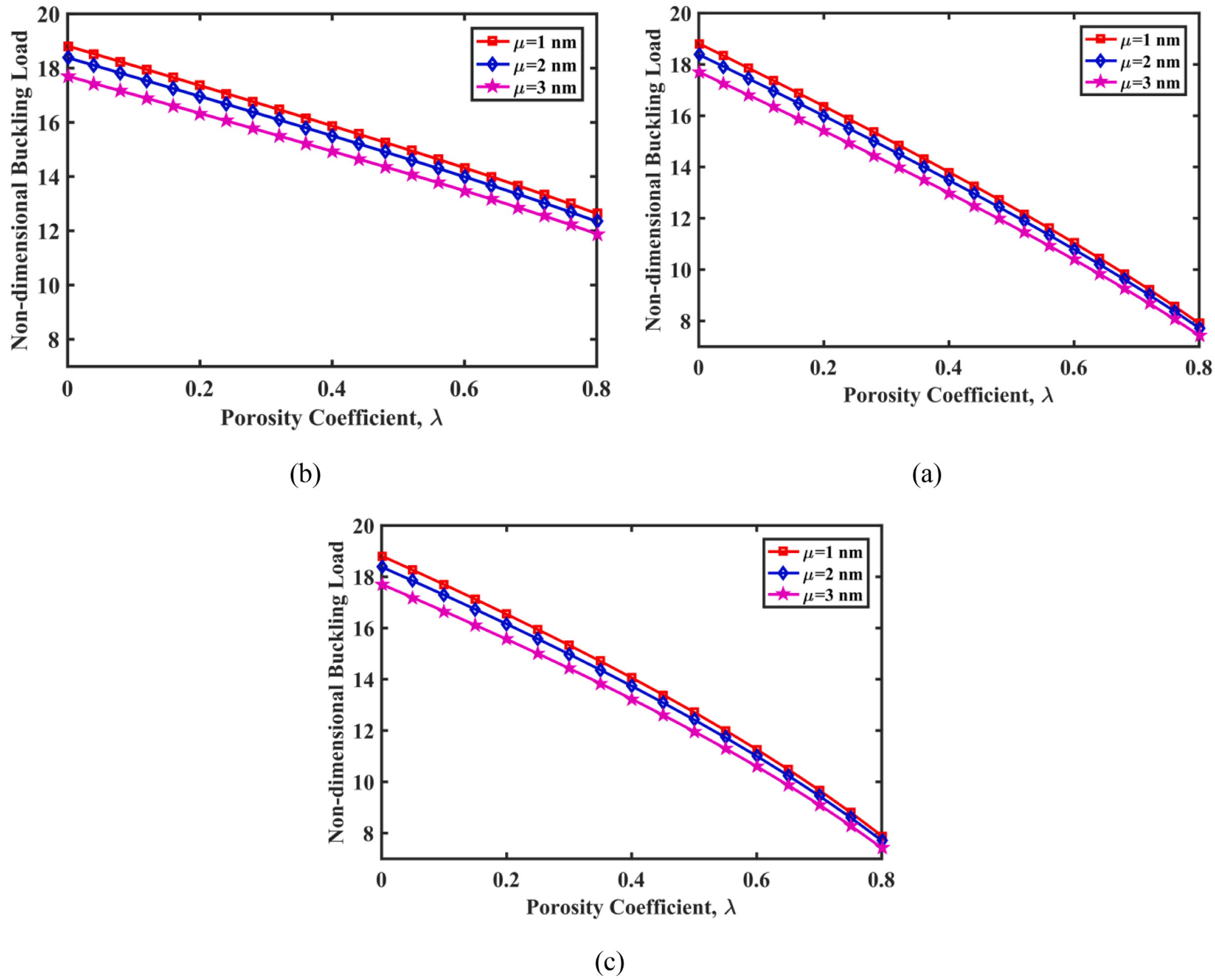

Fig. 7. Dimensionless CBL changes with Poros_Coef for different values of nl_parameter; (a) UD, (b) NUD1 and (c) NUD2.

$$\left\{\begin{matrix}\varepsilon_{xx}^{(0)}\\ \varepsilon_{yy}^{(0)}\\ \gamma_{xy}^{(0)}\end{matrix}\right\}=\left\{\begin{matrix}\dfrac{\partial u}{\partial x}\\ \dfrac{\partial v}{\partial y}\\ \dfrac{\partial u}{\partial y}+\dfrac{\partial v}{\partial x}\end{matrix}\right\},\ \left\{\begin{matrix}\kappa_{xx}^{b}\\ \kappa_{yy}^{b}\\ \kappa_{xy}^{b}\end{matrix}\right\}=\left\{\begin{matrix}-\dfrac{\partial^2 w_b}{\partial x^2}\\ -\dfrac{\partial^2 w_b}{\partial y^2}\\ -2\dfrac{\partial^2 w_b}{\partial x\partial y}\end{matrix}\right\},\ \left\{\begin{matrix}\kappa_{xx}^{s}\\ \kappa_{yy}^{s}\\ \kappa_{xy}^{s}\end{matrix}\right\}=\left\{\begin{matrix}-\dfrac{\partial^2 w_s}{\partial x^2}\\ -\dfrac{\partial^2 w_s}{\partial y^2}\\ -2\dfrac{\partial^2 w_s}{\partial x\partial y}\end{matrix}\right\},$$

$$\left\{\begin{matrix}\gamma_{yz}^{s}\\ \gamma_{xz}^{s}\end{matrix}\right\}=g(z)\left\{\begin{matrix}\dfrac{\partial w_s}{\partial y}\\ \dfrac{\partial w_s}{\partial x}\end{matrix}\right\},\ g(z)=1-\frac{d}{dz}f(z) \tag{16}$$

The following equation depicts the stress-strain relationship for FGP material [39]:

$$\left\{\begin{matrix}\sigma_{xx}\\ \sigma_{yy}\\ \sigma_{yz}\\ \sigma_{xz}\\ \sigma_{xy}\end{matrix}\right\}=\begin{bmatrix}C_{11} & C_{12} & 0 & 0 & 0\\ C_{12} & C_{22} & 0 & 0 & 0\\ 0 & 0 & C_{44} & 0 & 0\\ 0 & 0 & 0 & C_{55} & 0\\ 0 & 0 & 0 & 0 & C_{66}\end{bmatrix}\left\{\begin{matrix}\varepsilon_{xx}\\ \varepsilon_{yy}\\ \gamma_{yz}\\ \gamma_{xz}\\ \gamma_{xy}\end{matrix}\right\}. \tag{17}$$

Given that the material exhibits both isotropy and heterogeneity, the elastic constants in Eq. (17) can be expressed as:

$$\begin{aligned}&C_{11}=C_{22}=\frac{E(z)}{1-\nu^2},\\ &C_{12}=\frac{\nu E(z)}{1-\nu^2},\\ &C_{44}=C_{55}=C_{66}=G(z)=\frac{E(z)}{2(1+\nu)},\end{aligned} \tag{18}$$

where νrepresents Poisson's ratio. Given that the challenge is of nanoscale scale, it is required to think through the impact of size on mechanical properties. By employing the NSGT, it is possible to determine the structural equations of the nanoplate. The NSGT considers the strain gradient influence and the nonlocal stress field simultaneously through two-scale factors [44]. The differential form expression of the structural relationship of the NSGT is given by:

$$\begin{aligned}&\left[1-(e_1a)^2\nabla^2\right]\left[1-(e_0a)^2\nabla^2\right]\sigma_{ij}\\ &\quad=C_{ijkl}\left[1-(e_1a)^2\nabla^2\right]\varepsilon_{kl}-C_{ijkl}l^2\left[1-(e_0a)^2\nabla^2\right]\nabla^2\varepsilon_{kl}\end{aligned} \tag{19}$$

where C_{ijkl} is the elasticity tensor, e_0a and e_1a represent the nl_parameters, and l is the LS parameter of the material to include the strain gradient effects. Moreover, ∇^2 is the Laplacian operator. By applying $e_0 = e_1 = e$, the general form of Eq. (19) is simplified as follows [44]:

$$\left[1-(ea)^2\nabla^2\right]\sigma_{ij}=\left[1-l^2\nabla^2\right]C_{ijkl}\varepsilon_{kl} \tag{20}$$

Eq. (20) is the final differential form of NSGT. The influence of the NSGT is incorporated into the structural equation of the nanostructure's

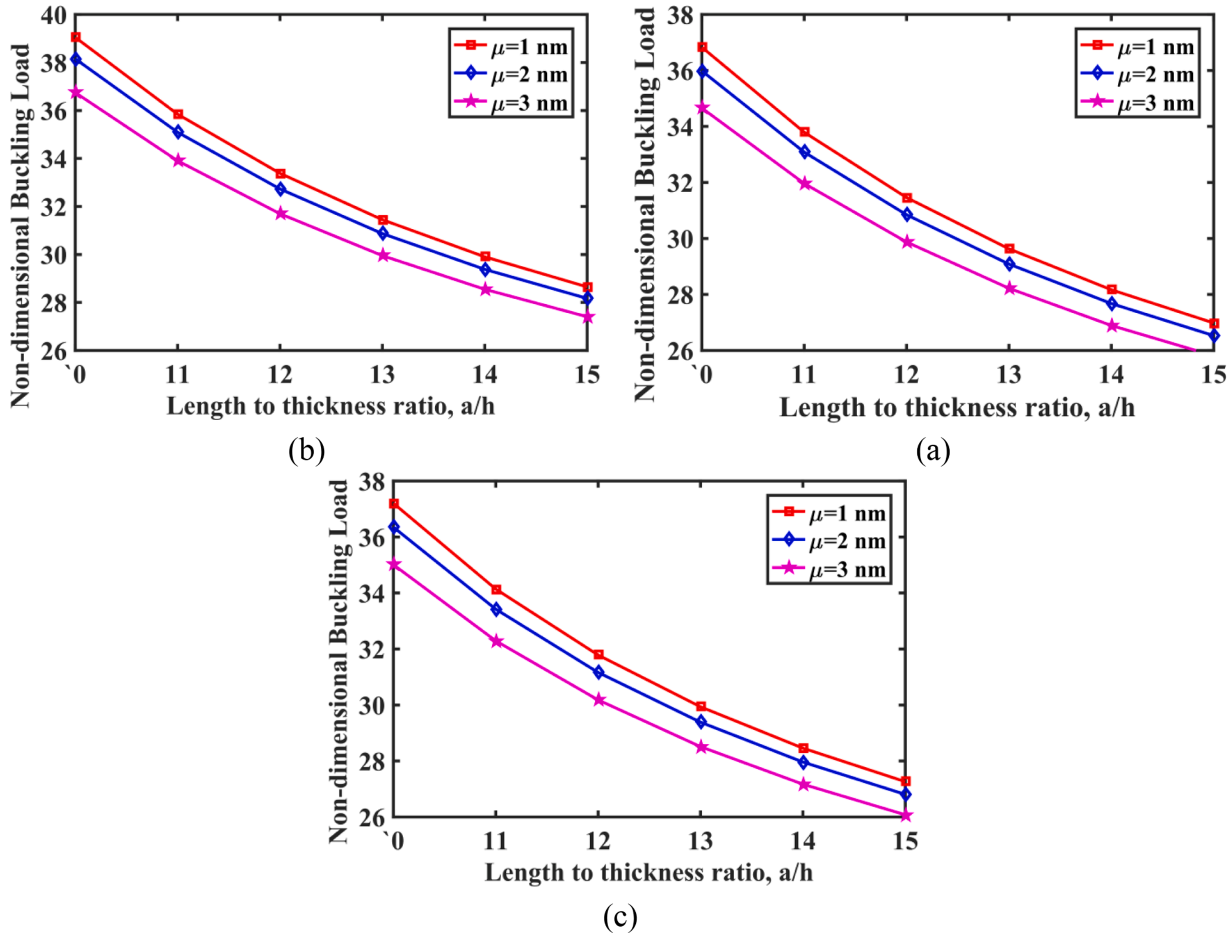

Fig. 8. Dimensionless CBL changes versus nanoplate length to thickness ratio for different values of nl_parameter; (a) UD, (b) NUD1 and (c) NUD2.

constituent material during the equation derivation procedure.

$$\left(1-\mu^2\nabla^2\right)\begin{Bmatrix}\sigma_{xx}\\ \sigma_{yy}\\ \sigma_{yz}\\ \sigma_{xz}\\ \sigma_{xy}\end{Bmatrix}=\left(1-l^2\nabla^2\right)\left(\begin{bmatrix}C_{11} & C_{12} & 0 & 0 & 0\\ C_{12} & C_{22} & 0 & 0 & 0\\ 0 & 0 & C_{44} & 0 & 0\\ 0 & 0 & 0 & C_{55} & 0\\ 0 & 0 & 0 & 0 & C_{66}\end{bmatrix}\begin{Bmatrix}\varepsilon_{xx}\\ \varepsilon_{yy}\\ \gamma_{yz}\\ \gamma_{xz}\\ \gamma_{xy}\end{Bmatrix}\right), \tag{21}$$

where $\mu = e\,a$ stands for the nl_parameter and l is the LS parameter.

The principle of minimal total potential energy is utilized to derive partial differential equations (PDEs) that govern the behavior of FGP nanoplate. As long as the system remains in static equilibrium, the total potential energy (comprising strain energy and work performed by external masses) will be minimal, in accordance with this principle. Eq. (22) expresses the principle governing the system's minimum total potential energy in light of this:

$$\delta(U+V_N+V_F)=0, \tag{22}$$

where δ is the operator of changes, U is the strain energy, and V_N and V_F represent the work done by in-plane loads and elastic media, respectively.

The virtual strain energy for the FGP nanoplate can be written as following:

$$\delta U=\int_A\int_{-h/2}^{h/2}\sigma_{xx}\,\delta\varepsilon_{xx}+\sigma_{yy}\,\delta\varepsilon_{yy}+\sigma_{xz}\,\delta\gamma_{xz}+\sigma_{yz}\,\delta\gamma_{yz}+\sigma_{xy}\,\delta\gamma_{xy}\Big)dz\,dA. \tag{23}$$

Eq. (23) is modified by substituting the strains from Eqs. (3-15) and (3–16) and utilizing the variational operator. This results in:

$$\begin{aligned}\delta U=&\int_A\int_{-h/2}^{h/2}\left(\sigma_{xx}\delta\left(\frac{\partial u}{\partial x}-z\frac{\partial^2 w_b}{\partial x^2}-f(z)\frac{\partial^2 w_s}{\partial x^2}\right)+\sigma_{yy}\delta\left(\frac{\partial v}{\partial y}-z\frac{\partial^2 w_b}{\partial y^2}-f(z)\frac{\partial^2 w_s}{\partial y^2}\right)\right.\\ &+\sigma_{xz}\delta\left(g(z)\frac{\partial w_s}{\partial x}\right)+\sigma_{yz}\delta\left(g(z)\frac{\partial w_s}{\partial y}\right)\\ &\left.+\sigma_{xy}\delta\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}-2z\frac{\partial^2 w_b}{\partial x\partial y}-2f(z)\frac{\partial^2 w_s}{\partial x\partial y}\right)\right)dz\,dA.\end{aligned} \tag{24}$$

where

$$\begin{aligned}&\left(N_{\alpha\beta},M^b_{\alpha\beta},M^s_{\alpha\beta}\right)=\int_{-h/2}^{h/2}\sigma_{\alpha\beta}(1,\,z,\,f(z))\,dz,\quad \alpha=x,y\quad \beta=x,y\\ &Q_{\alpha z}=\int_{-h/2}^{h/2}\sigma_{\alpha z}g(z)\,dz,\quad \alpha=x,y\end{aligned} \tag{25}$$

It should be mentioned that in Eq. (25), $N_{\alpha\beta}$ represents the resulting forces, $M^b_{\alpha\beta}$ and $M^s_{\alpha\beta}$ are the resulting moments, and $Q_{\alpha z}$ is the transverse shear forces.

What in-plane loads do to the nanoplate is [43]:

$$V_N=\frac{1}{2}\int_A\left(N^M_{xx}\left(\frac{\partial(w_b+w_s)}{\partial x}\right)^2+N^M_{yy}\left(\frac{\partial(w_b+w_s)}{\partial y}\right)^2\right)dA, \tag{26}$$

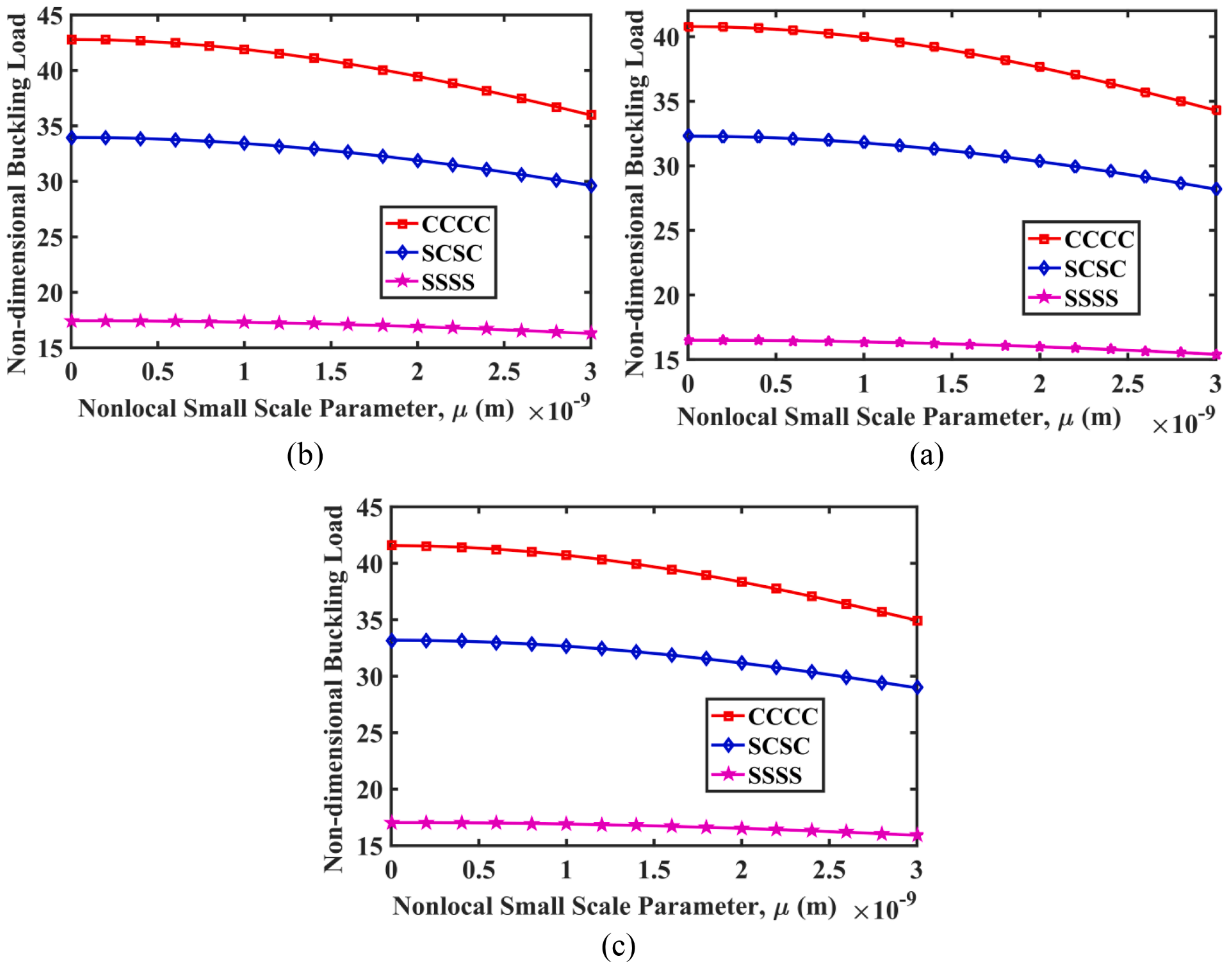

Fig. 9. Dimensionless CBL versus nl_parameter for different BCs; (a) UD, (b) NUD1 and (c) NUD2.

where N_{xx}^M and N_{yy}^M are in-plane loading functions. Eq. (27) represents the virtual work induced by in-plane loadings when the variational operator is applied to Eq. (26):

$$\delta V_N = \int_A \left(N_{xx}^M \frac{\partial(w_b + w_s)}{\partial x} \frac{\partial \delta(w_b + w_s)}{\partial x} + N_{yy}^M \frac{\partial(w_b + w_s)}{\partial y} \frac{\partial \delta(w_b + w_s)}{\partial y} \right) dA. \tag{27}$$

The nanoplate is exposed to both uniform and non-uniform compressive biaxial loads in this investigation. The functions of this form of loading are as follows:

$$N_{xx}^M = -P_0 \left(1 - \chi \left(\frac{y}{b} \right)^{1 \text{ or } 2} \right), \quad N_{yy}^M = -P_0 \left(1 - \chi \left(\frac{x}{a} \right)^{1 \text{ or } 2} \right), \tag{28}$$

where χ is the loading coefficient and P_0 represents the compressive force per length unit. The equations have a linear distribution when the loading function is of the first order, and a non-linear distribution when the second order is chosen. Fig. 2 displays the linear and non-linear loads examined in this study for various χ values.

The elastic substrate's work, as simulated by Pasternak's model, can be stated as Eq. (29):

$$V_F = -\frac{1}{2} \int_A \left[k_w (w_b + w_s) - k_s \nabla^2 (w_b + w_s) \right] (w_b + w_s) \, dA, \tag{29}$$

where k_g and k_w are Pasternak and Winkler constants of the elastic substrate. The application of the variational operator to the aforementioned equation yields the virtual work performed by the elastic substrate.

$$\delta V_F = \int_A \left[-k_w (w_b + w_s) \delta(w_b + w_s) + k_g \left(\frac{\partial(w_b + w_s)}{\partial x} \frac{\partial \delta(w_b + w_s)}{\partial x} + \frac{\partial(w_b + w_s)}{\partial y} \frac{\partial \delta(w_b + w_s)}{\partial y} \right) \right] dA. \tag{30}$$

The subsequent steps involve substituting Eqs. (24), (27), and (30) into Eq. (22), performing part integration, and equating the coefficients of δu,δv, δw_b and δw_s as zero to arise the following Euler-Lagrange equations:

$$\frac{\partial N_{xx}}{\partial x} + \frac{\partial N_{xy}}{\partial y} = 0, \tag{31}$$

$$\frac{\partial N_{xy}}{\partial x} + \frac{\partial N_{yy}}{\partial y} = 0, \tag{32}$$

$$\frac{\partial^2 M_{xx}^b}{\partial x^2} + 2 \frac{\partial^2 M_{xy}^b}{\partial x \partial y} + \frac{\partial^2 M_{yy}^b}{\partial y^2} + N_{xx}^M \frac{\partial^2 (w_b + w_s)}{\partial x^2} + N_{yy}^M \frac{\partial^2 (w_b + w_s)}{\partial y^2} - k_w (w_b + w_s) + k_g \left(\frac{\partial^2 (w_b + w_s)}{\partial x^2} + \frac{\partial^2 (w_b + w_s)}{\partial y^2} \right) = 0, \tag{33}$$

$$\frac{\partial^2 M_{xx}^s}{\partial x^2} + 2 \frac{\partial^2 M_{xy}^s}{\partial x \partial y} + \frac{\partial^2 M_{yy}^s}{\partial y^2} + \frac{\partial Q_{xz}}{\partial x} + \frac{\partial Q_{yz}}{\partial y} + N_{xx}^M \frac{\partial^2 (w_b + w_s)}{\partial x^2} + N_{yy}^M \frac{\partial^2 (w_b + w_s)}{\partial y^2} - k_w (w_b + w_s) + k_g \left(\frac{\partial^2 (w_b + w_s)}{\partial x^2} + \frac{\partial^2 (w_b + w_s)}{\partial y^2} \right) = 0. \tag{34}$$

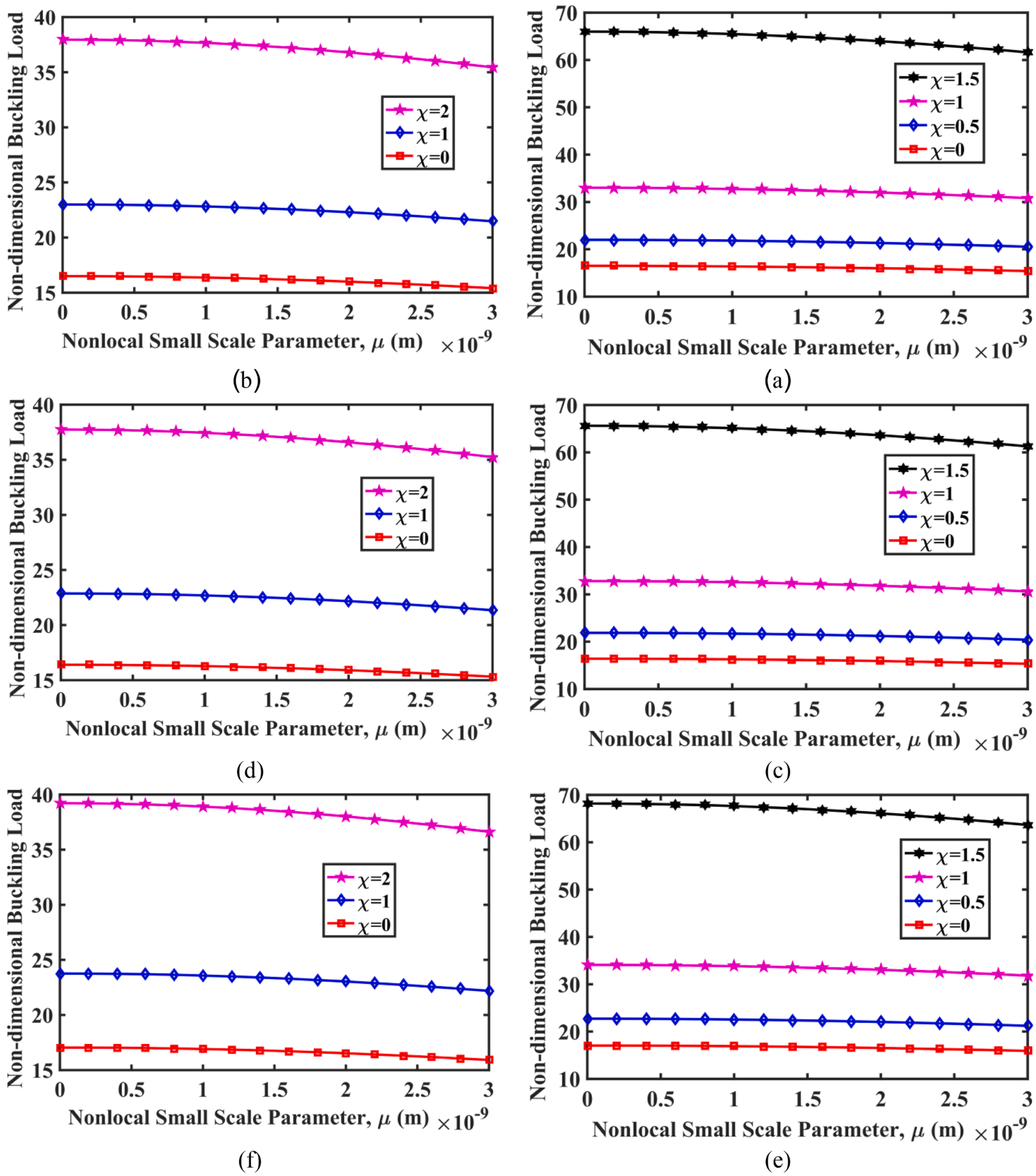

Fig. 10. Dimensionless CBL changes with nl_parameter for (a,c,e) Linear in-plane loading and (b,d,f) Non-linear in-plane loading; (a,b) UD, (c,d) NUD1 and (e, f) NUD2.

The BCs of the nanoplate in this work are regarded as simply supported and clamped. The aforementioned BCs will result in the application of the subsequent conditions on the edges of the plate [45]:

- Clamped supports

$$\text{at } (x=0,a \text{ and } y=0,b) \Rightarrow u=v=w_b=w_s=\frac{\partial w_b}{\partial x}=\frac{\partial w_b}{\partial y}=\frac{\partial w_s}{\partial x}=\frac{\partial w_s}{\partial y}0, \tag{35}$$

- Simply supported edges

$$\begin{aligned} &\text{at } x=0,a \Rightarrow v=w_b=w_s=N_{xx}=M_{xx}^b=M_{xx}^s=0, \\ &\text{at } y=0,b \Rightarrow u=w_b=w_s=N_{yy}=M_{yy}^b=M_{yy}^s=0. \end{aligned} \tag{36}$$

In order to determine the resultant forces and moments using the NSGT, Eq. (21) is inserted into Eq. (25). The outcome is as follows:

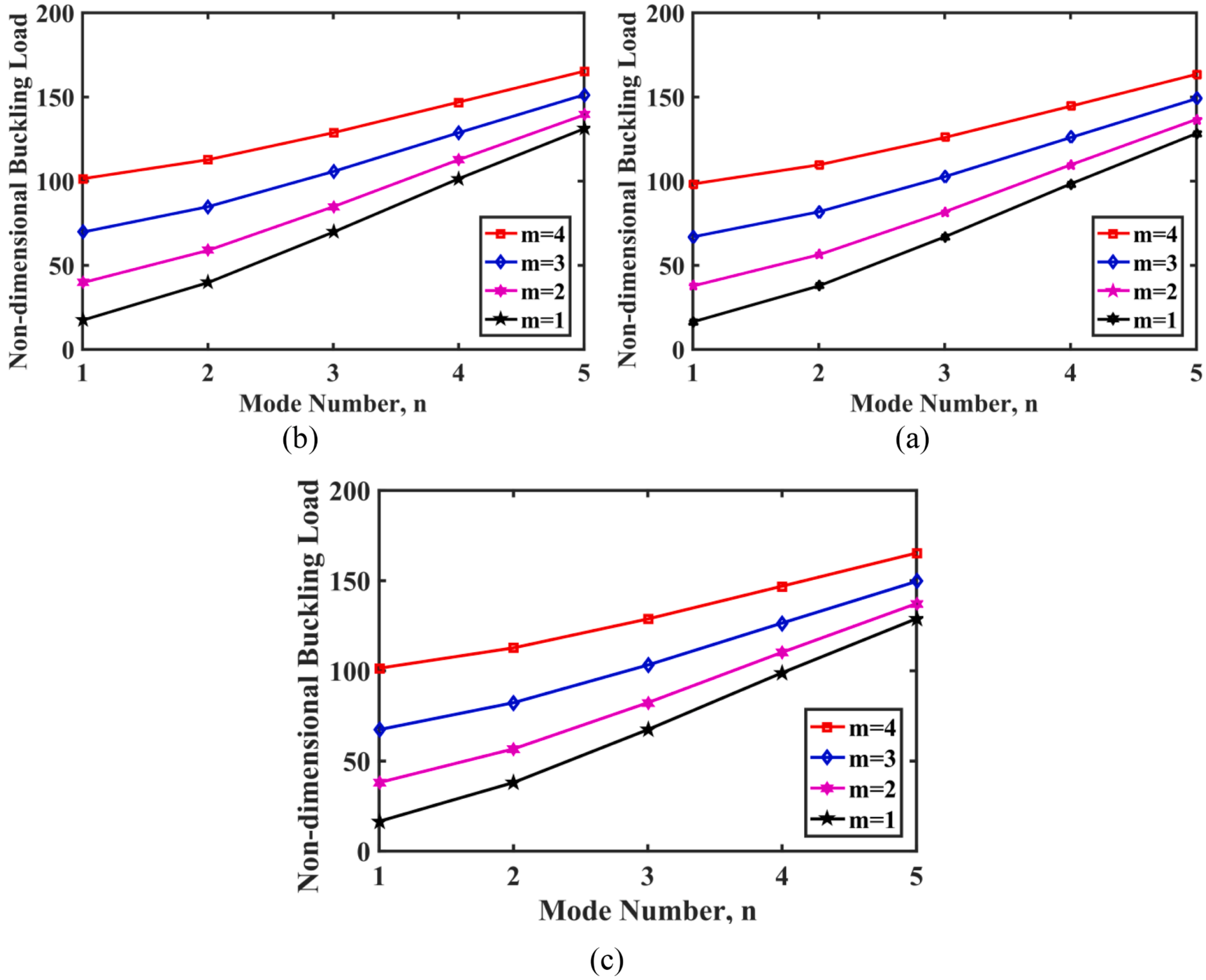

Fig. 11. Dimensionless CBL changes with buckling mode number n; (a) UD, (b) NUD1 and (c) NUD2.

$$
\left(1-\mu^2\nabla^2\right)\begin{Bmatrix} N_{xx} \\ N_{xx} \\ N_{xy} \end{Bmatrix} = \left(1-l^2\nabla^2\right)\left(\begin{bmatrix} A_{11} & A_{12} & 0 \\ A_{12} & A_{22} & 0 \\ 0 & 0 & A_{66} \end{bmatrix}\begin{Bmatrix} \dfrac{\partial u}{\partial x} \\ \dfrac{\partial v}{\partial y} \\ \dfrac{\partial u}{\partial y}+\dfrac{\partial v}{\partial x} \end{Bmatrix} + \begin{bmatrix} B_{11} & B_{12} & 0 \\ B_{12} & B_{22} & 0 \\ 0 & 0 & B_{66} \end{bmatrix}\begin{Bmatrix} -\dfrac{\partial^2 w_b}{\partial x^2} \\ -\dfrac{\partial^2 w_b}{\partial y^2} \\ -2\dfrac{\partial^2 w_b}{\partial x \partial y} \end{Bmatrix} + \begin{bmatrix} B^s_{11} & B^s_{12} & 0 \\ B^s_{12} & B^s_{22} & 0 \\ 0 & 0 & B^s_{66} \end{bmatrix}\begin{Bmatrix} -\dfrac{\partial^2 w_s}{\partial x^2} \\ -\dfrac{\partial^2 w_s}{\partial y^2} \\ -2\dfrac{\partial^2 w_s}{\partial x \partial y} \end{Bmatrix}\right), \tag{37}
$$

$$
\left(1-\mu^2\nabla^2\right)\begin{Bmatrix} M^b_{xx} \\ M^b_{yy} \\ M^b_{xy} \end{Bmatrix} = \left(1-l^2\nabla^2\right)\left(\begin{bmatrix} B_{11} & B_{12} & 0 \\ B_{12} & B_{22} & 0 \\ 0 & 0 & B_{66} \end{bmatrix}\begin{Bmatrix} \dfrac{\partial u}{\partial x} \\ \dfrac{\partial v}{\partial y} \\ \dfrac{\partial u}{\partial y}+\dfrac{\partial v}{\partial x} \end{Bmatrix} + \begin{bmatrix} D_{11} & D_{12} & 0 \\ D_{12} & D_{22} & 0 \\ 0 & 0 & D_{66} \end{bmatrix}\begin{Bmatrix} -\dfrac{\partial^2 w_b}{\partial x^2} \\ -\dfrac{\partial^2 w_b}{\partial y^2} \\ -2\dfrac{\partial^2 w_b}{\partial x \partial y} \end{Bmatrix} + \begin{bmatrix} D^s_{11} & D^s_{12} & 0 \\ D^s_{12} & D^s_{22} & 0 \\ 0 & 0 & D^s_{66} \end{bmatrix}\begin{Bmatrix} -\dfrac{\partial^2 w_s}{\partial x^2} \\ -\dfrac{\partial^2 w_s}{\partial y^2} \\ -2\dfrac{\partial^2 w_s}{\partial x \partial y} \end{Bmatrix}\right), \tag{38}
$$

$$(1-\mu^2\nabla^2)\begin{Bmatrix} M_{xx}^s \\ M_{yy}^s \\ M_{xy}^s \end{Bmatrix} = (1-l^2\nabla^2)\left(\begin{bmatrix} B_{11}^s & B_{12}^s & 0 \\ B_{12}^s & B_{22}^s & 0 \\ 0 & 0 & B_{66}^s \end{bmatrix}\begin{Bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y}+\frac{\partial v}{\partial x} \end{Bmatrix}\right.$$
$$\left.+\begin{bmatrix} D_{11}^s & D_{12}^s & 0 \\ D_{12}^s & D_{22}^s & 0 \\ 0 & 0 & D_{66}^s \end{bmatrix}\begin{Bmatrix} -\frac{\partial^2 w_b}{\partial x^2} \\ -\frac{\partial^2 w_b}{\partial y^2} \\ -2\frac{\partial^2 w_b}{\partial x \partial y} \end{Bmatrix} + \begin{bmatrix} H_{11}^s & H_{12}^s & 0 \\ H_{12}^s & H_{22}^s & 0 \\ 0 & 0 & H_{66}^s \end{bmatrix}\begin{Bmatrix} -\frac{\partial^2 w_s}{\partial x^2} \\ -\frac{\partial^2 w_s}{\partial y^2} \\ -2\frac{\partial^2 w_s}{\partial x \partial y} \end{Bmatrix}\right), \tag{39}$$

$$(1-\mu^2\nabla^2)\begin{Bmatrix} Q_{xz} \\ Q_{yz} \end{Bmatrix} = (1-l^2\nabla^2)\left(\begin{bmatrix} A_{55}^s & 0 \\ 0 & A_{44}^s \end{bmatrix}\begin{Bmatrix} \frac{\partial w_s}{\partial x} \\ \frac{\partial w_s}{\partial y} \end{Bmatrix}\right), \tag{40}$$

where

$$\begin{bmatrix} A_{11} & B_{11} & D_{11} & B_{11}^s & D_{11}^s & H_{11}^s \\ A_{12} & B_{12} & D_{12} & B_{12}^s & D_{12}^s & H_{12}^s \\ A_{22} & B_{22} & D_{22} & B_{22}^s & D_{22}^s & H_{22}^s \\ A_{66} & B_{66} & D_{66} & B_{66}^s & D_{66}^s & H_{66}^s \end{bmatrix} = \int_{-h/2}^{h/2} \begin{Bmatrix} C_{11} \\ C_{12} \\ C_{22} \\ C_{66} \end{Bmatrix}\{\, 1 \quad z \quad z^2 \quad f(z) \quad zf(z) \quad f^2(z) \,\}dz,$$

$$\{\, A_{44}^s \quad A_{55}^s \,\} = \int_{-h/2}^{h/2} \{\, C_{44} \quad C_{55} \,\} g^2(z)\,dz. \tag{41}$$

In closing, the PDEs governing the FGP nanoplate situated on the Pasternak medium and founded on the NSGT are deduced by substituting the resulting forces and torques into Eqs. (31) to (34):

$$(1-l^2\nabla^2)\left\{A_{11}\frac{\partial^2 u}{\partial x^2}+A_{12}\frac{\partial^2 v}{\partial x \partial y}-B_{11}\frac{\partial^3 w_b}{\partial x^3}-B_{12}\frac{\partial^3 w_b}{\partial x \partial y^2}-B_{11}^s\frac{\partial^3 w_s}{\partial x^3}-B_{12}^s\frac{\partial^3 w_s}{\partial x \partial y^2}\right.$$
$$\left.+A_{66}\left(\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 v}{\partial x \partial y}\right)-2B_{66}\frac{\partial^3 w_b}{\partial x \partial y^2}-2B_{66}^s\frac{\partial^3 w_s}{\partial x \partial y^2}\right\}=0, \tag{42}$$

$$(1-l^2\nabla^2)\left\{A_{22}\frac{\partial^2 v}{\partial y^2}+A_{12}\frac{\partial^2 u}{\partial x \partial y}-B_{22}\frac{\partial^3 w_b}{\partial y^3}-B_{12}\frac{\partial^3 w_b}{\partial x^2 \partial y}-B_{22}^s\frac{\partial^3 w_s}{\partial y^3}-B_{12}^s\frac{\partial^3 w_s}{\partial x^2 \partial y}\right.$$
$$\left.+A_{66}\left(\frac{\partial^2 u}{\partial x \partial y}+\frac{\partial^2 v}{\partial x^2}\right)-2B_{66}\frac{\partial^3 w_b}{\partial x^2 \partial y}-2B_{66}^s\frac{\partial^3 w_s}{\partial x^2 \partial y}\right\}=0, \tag{43}$$

$$(1-l^2\nabla^2)\left\{B_{11}\frac{\partial^3 u}{\partial x^3}+B_{12}\frac{\partial^3 v}{\partial x^2 \partial y}-D_{11}\frac{\partial^4 w_b}{\partial x^4}-D_{11}^s\frac{\partial^4 w_s}{\partial x^4}+2B_{66}\left(\frac{\partial^3 u}{\partial x \partial y^2}+\frac{\partial^3 v}{\partial x^2 \partial y}\right)\right.$$
$$-4D_{66}\frac{\partial^4 w_b}{\partial x^2 \partial y^2}-4D_{66}^s\frac{\partial^4 w_s}{\partial x^2 \partial y^2}+B_{12}\frac{\partial^3 u}{\partial x \partial y^2}+B_{22}\frac{\partial^3 v}{\partial y^3}-2D_{12}\frac{\partial^4 w_b}{\partial x^2 \partial y^2}$$
$$\left.-D_{22}\frac{\partial^4 w_b}{\partial y^4}-2D_{12}^s\frac{\partial^4 w_s}{\partial x^2 \partial y^2}-D_{22}^s\frac{\partial^4 w_s}{\partial y^4}\right\}$$
$$+(1-\mu^2\nabla^2)\left[N_{xx}^M\frac{\partial^2 (w_b+w_s)}{\partial x^2}+N_{yy}^M\frac{\partial^2 (w_b+w_s)}{\partial y^2}-k_w(w_b+w_s)\right.$$
$$\left.+k_g\left(\frac{\partial^2 (w_b+w_s)}{\partial x^2}+\frac{\partial^2 (w_b+w_s)}{\partial y^2}\right)\right]=0, \tag{44}$$

$$(1-l^2\nabla^2)\left\{B_{11}^s\frac{\partial^3 u}{\partial x^3}+B_{12}^s\frac{\partial^3 v}{\partial x^2 \partial y}-D_{11}^s\frac{\partial^4 w_b}{\partial x^4}-H_{11}^s\frac{\partial^4 w_s}{\partial x^4}+2B_{66}^s\left(\frac{\partial^3 u}{\partial x \partial y^2}+\frac{\partial^3 v}{\partial x^2 \partial y}\right)\right.$$
$$-4D_{66}^s\frac{\partial^4 w_b}{\partial x^2 \partial y^2}-4H_{66}^s\frac{\partial^4 w_s}{\partial x^2 \partial y^2}+B_{12}^s\frac{\partial^3 u}{\partial x \partial y^2}+B_{22}^s\frac{\partial^3 v}{\partial y^3}-2D_{12}^s\frac{\partial^4 w_b}{\partial x^2 \partial y^2}$$
$$\left.-D_{22}^s\frac{\partial^4 w_b}{\partial y^4}-2H_{12}^s\frac{\partial^4 w_s}{\partial x^2 \partial y^2}-H_{22}^s\frac{\partial^4 w_s}{\partial y^4}+A_{55}^s\frac{\partial^2 w_s}{\partial x^2}+A_{44}^s\frac{\partial^2 w_s}{\partial y^2}\right\}$$
$$+(1-\mu^2\nabla^2)\left[N_{xx}^M\frac{\partial^2 (w_b+w_s)}{\partial x^2}+N_{yy}^M\frac{\partial^2 (w_b+w_s)}{\partial y^2}-k_w(w_b+w_s)\right.$$
$$\left.+k_g\left(\frac{\partial^2 (w_b+w_s)}{\partial x^2}+\frac{\partial^2 (w_b+w_s)}{\partial y^2}\right)\right]=0. \tag{45}$$

The coupled partial differential Eqs. (42) to (45) govern the static equilibrium of an FGP nanoplate on an elastic medium, taking into consideration the nonlocal strain gradient.

3. Problem solution

For various BCs, the Galerkin approach is employed to solve these equations. Accordingly, the functions $\{u, v, w_b, w_s\}$ are written using trigonometric functions as follows:

$$u(x,y) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} U_{mn}\frac{\partial F_m(x)}{\partial x}G_n(y), \tag{46}$$

$$u(x,y) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} U_{mn}\frac{\partial F_m(x)}{\partial x}G_n(y),$$

$$w_b(x,y) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} W_{bmn}F_m(x)G_n(y),$$

$$w_s(x,y) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} W_{smn}F_m(x)G_n(y),$$

where $\{U_{mn}, V_{mn}, W_{bmn}, W_{smn}\}$ are unknown coefficients and m and n are mode numbers. The selection of shape functions $F_m(x)$ and $G_n(y)$ is based on their ability to meet the requirements of the problem's BCs. These functions for different BCs are presented below [45]:

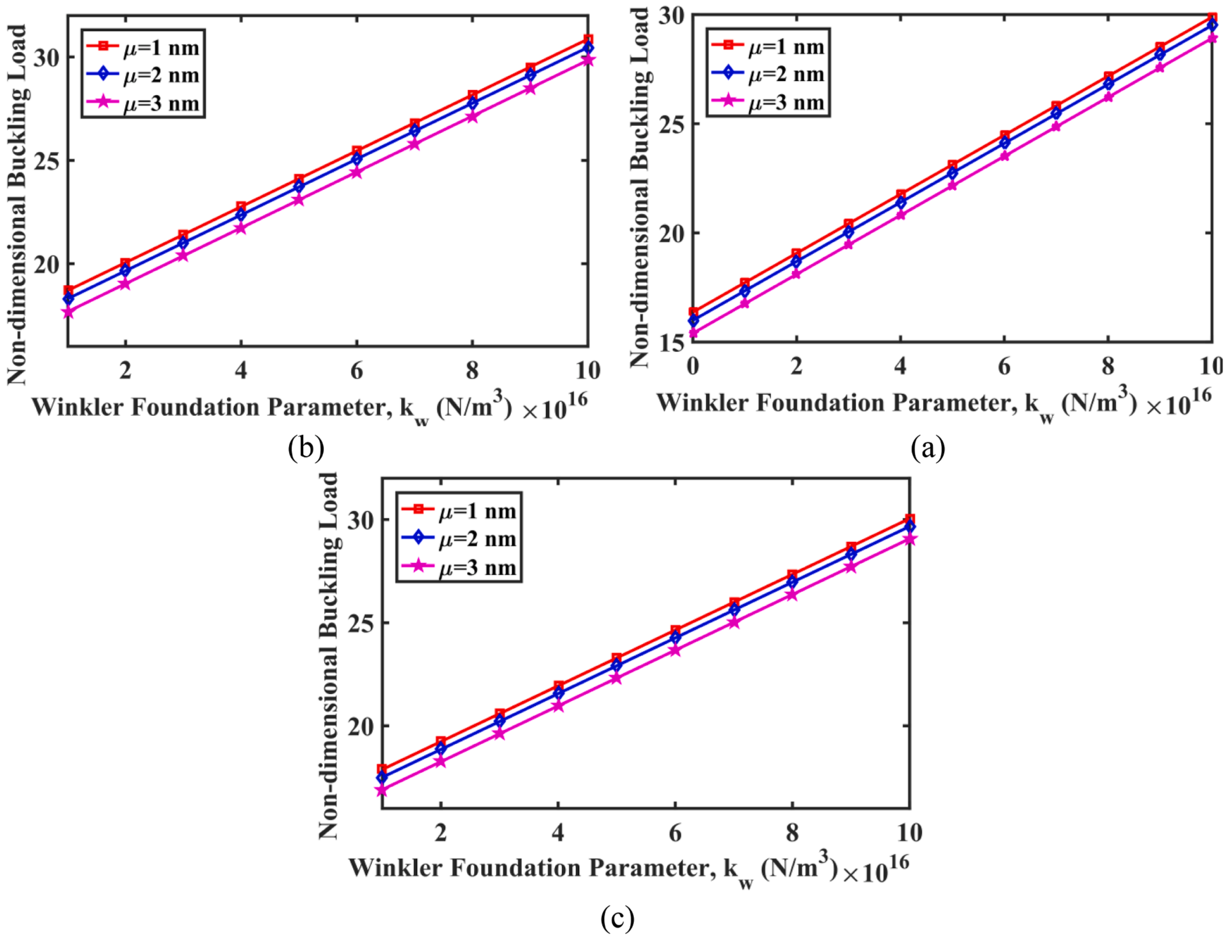

Fig. 12. Dimensionless CBL changes versus the Winkler parameter for different values of the nl_parameter; (a) UD, (b) NUD1 and (c) NUD2.

- SSSS supports

$$F_m(x) = \sin\left(\frac{m\pi x}{a}\right), \quad G_n(y) = \sin\left(\frac{n\pi y}{b}\right), \tag{47}$$

- CCCC supports

$$F_m(x) = \sin^2\left(\frac{m\pi x}{a}\right), \quad G_n(y) = \sin^2\left(\frac{n\pi y}{b}\right), \tag{48}$$

- SCSC supports

$$F_m(x) = \sin\left(\frac{m\pi x}{a}\right)\left[\cos\left(\frac{m\pi x}{a}\right) - 1\right], \ G_n(y) = \sin\left(\frac{n\pi y}{b}\right)\left[\cos\left(\frac{n\pi y}{b}\right) - 1\right]. \tag{49}$$

By substituting Eq. (46) into Eqs. (42) to (45) and using the Galerkin technique, the following algebraic equations are obtained:

$$\begin{bmatrix} K_{11} & K_{12} & K_{13} & K_{14} \\ K_{21} & K_{22} & K_{23} & K_{24} \\ K_{31} & K_{32} & K_{33} & K_{34} \\ K_{41} & K_{42} & K_{43} & K_{44} \end{bmatrix} \begin{Bmatrix} U_{mn} \\ V_{mn} \\ W_{bmn} \\ W_{smn} \end{Bmatrix} = 0. \tag{50}$$

To calculate the CBL, the determinant of the matrix of coefficients (stiffness matrix) is calculated and set equal to zero. By solving the resulting equation for P_0, the CBL is obtained.

4. Validation study

In this step, the extracted formulation is validated before presenting and analyzing the results. To achieve this objective, the CBL is juxtaposed with the findings of Narandar's study [46]. The critical load of a graphene nanoplate with simple supports was determined in Narandar's study by the use of the corrected plate theory, Eringen's nonlocal theory, and Navier's method. We modify the mechanical qualities and geometric attributes in accordance with reference [46]. Fig. 3 displays the CBL ratio estimated using the formulation developed in this study and the results obtained from Narendar's study [46], as a function of the nl_parameter. Eq. (51) defines the CBL ratio as the proportion between the buckling loads determined using the nonlocal theory and the local theory.

$$\text{BucklingLoadRatio} = \frac{\text{Buckling loadcalculated using nonlocaltheory}}{\text{Buckling loadcalculated using localtheory}}. \tag{51}$$

As shown in Fig. 3, the obtained results and those of Narendar's study for various nanoplate dimensions are in excellent accord. Fig. 4 compares, for additional substantiation, the CBL ratio for various buckling mode numbers as a function of the nl_parameter. As evidenced by the results' high degree of consistency, the current formulation is accurate.

5. Results and discussions

This section analyzes the impact of various parameters on the CBL of the FGP nanoplate. The nanoplate is made of metal foam with $E_0 = 200$ GPa, $G_0 = 76.92$ GPa and $\nu = 1/3$ as considered in Ref. [39]. For

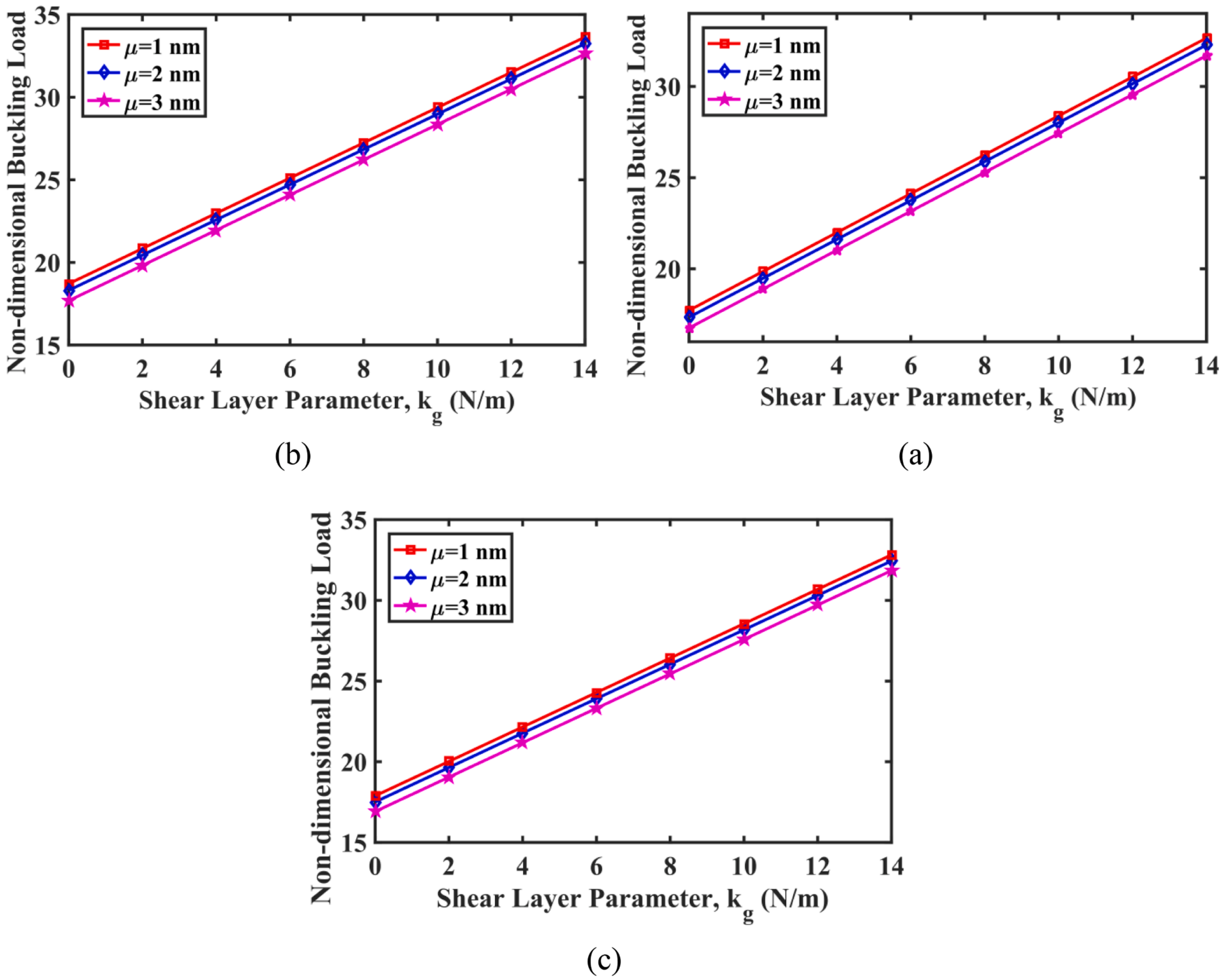

Fig. 13. Dimensionless CBL variations with shear layer stiffness for different values of nl_parameter; (a) UD, (b) NUD1 and (c) NUD2.

convenience, the dimensionless CBL is defined as follows:

$$\overline{P} = P_0 \times \frac{a^2}{D}; \quad D = \frac{E_0 h^3}{12(1-\nu^2)} \tag{52}$$

For different values of the LS parameter and porosity distributions, Fig. 5 illustrates the relationship between the dimensionless CBL of the metal foam nanoplate and the nl_parameter. The nanoplate, with simple supports and dimensions $a = b = 50$ nm and thickness $h = 5$ nm, is assumed to be subjected to an uniform biaxial compressive in-plane load, irrespective of the elastic substrate. The observed trend shows that when the nl_parameter grows, the dimensionless CBL for different porosity distributions decreases significantly. This can be attributed to the fact that an increase in the nl_parameter results in an increase in the softness of the nanostructure. Furthermore, it can be observed from Fig. 5 that when the LS parameter increases, there is a corresponding increase in the dimensionless CBL. The rationale behind this is that augmenting the LS parameter will result in an elevation of the nanoplate's stiffness.

The impact of the porosity distribution type on the dimensionless CBL is seen in Fig. 6. This analysis examines two distinct categories of nanoplates, namely square and rectangular, subjected to uniform biaxial in-plane loading. A square nanoplate with length and width $a = b =$ 50 nm and a rectangular nanoplate with length of $a = 50$ nm and width of $b = 75$ nm are assumed. System parameters are set as $k_w = 0$, $k_g = 0$, $m = n = 1$, $l = 1$ nmand $\lambda = 0.2$. It is evident that irrespective of the shape of the nanoplate, whether it is square or rectangular, the nanoplate exhibiting uniform porosity (UD) demonstrates more instability compared to the nanoplate with non-uniform porosity. This is because materials with uniform porosity tend to be less stiff than those with non-uniform porosity. Non-uniformity may provide stress distribution paths, improving the material's ability to withstand deformation. The dimensionless CBL is greater in the scenario where the porosity distribution is UD1 compared to the NUD2. Furthermore, Fig. 6 demonstrates that the square nanoplate exhibits far greater stability than the rectangular nanoplate across all three phases of porosity distribution. This is due to the fact that, compared to square nanoplates, rectangle ones are less stiff.

Fig. 7 illustrates the effect of the Poros_Coef on the dimensionless CBL of a FGP nanoplate under uniform biaxial in-plane loading with simple supports, considering different porosity distributions. In this exploration, the system parameters are selected as $k_w = 0$, $k_g = 0$, $m = n = 1$, $l = 1$ nm, $a = b = 50$ nm and $h = 5$ nm. It is evident that across all three forms of porosity distribution, an upward trend in the Poros_Coef is accompanied by a notable decline in the dimensionless buckling load. This happens because raising the Poros_Coef increases the amount of void space, which reduces the nanoplate's capacity to distribute stress and resist deformation. Furthermore, based on the findings presented in Fig. 7, it can be inferred that the stability of the nanoplate with a NUD1 is relatively unaffected by the Poros_Coef. Conversely, the CBL of the nanoplate with both uniform and NUD2 is primarily influenced by the coefficient associated with porosity.

Fig. 8 illustrates the relationship between the dimensionless CBL and the ratio of the length to the thickness of the porous nanoplate, at various values of the nl_parameter. In this analysis, a square nanoplate with dimensions $a = b = 50$ nm with uniform porosity and Poros_Coef

$\lambda = 0.2$ is considered. The system parameters are assumed as $k_w = 0$, $k_g = 0$, $m = n = 1$ and $l = 1$ nm. An increase in the length to thickness ratio results in a reduction of the dimensionless CBL. This is due to the fact that increasing the length to thickness ratio enhances the softness of the nanoplate. Furthermore, as the nl_parameter rises, there is an associated drop in the dimensionless CBL.

The impact of BCs on dimensionless CBL is illustrated in Fig. 9. The present investigation involves the application of biaxial and in-plane compressive loads on a square nanoplate characterized by uniform porosity. Ignoring the elastic medium, the system parameters are assumed as $a = b = 50$ nm, $h = 5$ nm, $m = n = 1$, $l = 1$ nm and $\lambda = 0.2$. There are three distinct sorts of BCs that have been incorporated for the edges of the nanoplate. In the first scenario, all edges of the nanoplate are equipped with simple supports. In the second scenario, the nanoplate exhibits fixed edges, but in the third scenario, two edges of the nanoplate are fixed while the remaining two edges are simply supported. The findings indicate that the dimensionless CBL is significantly affected by the BCs. Specifically, the static stability of the FGP nanoplate is found to be larger in the state where all of its edges are clamped compared to other states. The underlying cause of this behavior can be attributed to the clamped supports, which serve to enhance the stiffness of the nanoplate by imposing additional constraints on its edges. Furthermore, as depicted in Fig. 9, simply supported nanoplates have worse static stability in comparison to alternative scenarios. The stated findings are valid across various porosity distributions.

Fig. 10 examines the impact of various in-plane stress circumstances on the buckling properties of FGP nanoplates, taking into account several nl_parameters and different porosity distributions. The variation of the dimensionless CBL versus nl_parameter for linear and non-linear in-plane loads are depicted in Figs. 10 (a) and (b), respectively. The dimensionless CBL grows as the nl_parameter rises for all values of the parameter χ. Furthermore, when comparing the dimensionless CBL in linear and non-linear modes, it is evident that the nanoplate exhibits reduced stability under non-linear pressure conditions.

In Fig. 11, the variations in dimensionless CBL are depicted about the buckling mode number n for an FGP nanoplate. The current investigation considers a square nanoplate with diverse porosity distributions that is simply supported, disregarding the elastic substrate. Increasing the number of buckling modes will result in an escalation in the dimensionless CBL. Furthermore, as the number of buckling modes (n) increases, the disparity in the outcomes associated with various buckling modes (m) diminishes.

Figs. 12 and 13 illustrate the variations in the dimensionless CBL as a function of the spring stiffness of the elastic substrate and the stiffness of the shear layer, respectively. It is clear that, regardless of the kind of porosity distribution, increasing the stiffness of the elastic foundation results in an increase in the CBL. In the results presented in Fig. 12, the parameter k_g is considered equal to zero. Therefore, in this particular scenario, the elastic foundation is represented by a Winkler substrate. On the other hand, in Fig. 13, by choosing non-zero values for the parameters k_g and k_w, the elastic medium is simulated by the Pasternak media. By comparing Figs. 12 and 13, it is seen that the dimensionless CBL in the presence of Pasternak media is higher than the dimensionless CBL in the presence of Winkler substrate. The reason for this is that Winkler substrate only considers vertical stresses, but Pasternak foundation also considers transverse shear stresses in addition to vertical stresses.

6. Conclusion

The objective of this study was to examine the buckling issue of FGP nanoplates subjected to both uniform and non-uniform in-plane loadings. Size effects were implemented following the principles of the NSGT. Furthermore, the nanoplate was computationally represented using the four-variable refined theory. To replicate the elastic conditions around the nanoplate, the Pasternak substrate model was employed, comprising a shear layer and a spring layer. The nanoplate was subjected to several in-plane compressive loads, both linear and non-linear, under assumed conditions. The Galerkin method was employed to solve the governing PDEs under various BCs, resulting in the extraction of the governing algebraic equations.

- The dimensionless CBL of the FGP nanoplate drops considerably as the nl_parameter increases because it softens the nanostructure. The dimensionless CBL of the FGP nanoplate grows with the LS parameter. Because raising the LS parameter makes the nanoplate harder.
- Nanoplates exhibiting uniform porosity exhibit lower stability in comparison to nanoplates with non-uniform porosity. The dimensionless CBL is higher in the case of non-uniform porosity distribution compared to the dimensionless CBL in the case of uniform porosity distribution. Furthermore, across all three porosity distribution scenarios, square nanoplates demonstrate greater stability than rectangular nanoplates.
- In all three examined porosity distributions, it is perceived that as the Poros_Coef increases, there is a notable decline in the dimensionless buckling load. This can be attributed to the fact that an increase in the Poros_Coef leads to a drop in the hardness of the nanoplate. Furthermore, the findings indicate that the variation in Poros_Coef has a lesser impact on the stability of nanoplates with non-uniform porosity distribution in the first type. Conversely, the Poros_Coef has the greatest influence on the buckling critical loads of nanoplates with both uniform and non-uniform porosity distribution in the second type.
- The findings indicate that the CBL is significantly influenced by the dimensions of the nanoplate. Specifically, as the ratio of the length to the thickness of the nanoplate increases, there is a substantial increase in the dimensionless buckling load.
- BCs significantly influence dimensionless buckling load, with FGP nanoplates having superior static stability when all edges are clamped. This happens because the supporting supports constrain the nanoplate's edges, making it harder. Furthermore, the static stability of a simply supported nanoplate is lower in comparison to other scenarios.
- The results related to the effect of in-plane compressive loading type show that for loading types with linear and non-linear distribution, dimensionless buckling increases with increasing load parameter χ. In addition, in the same parameter χ with the comparison of dimensionless CBL in linear and non-linear modes, it can be observed that the nanoplate has less static stability when it is subjected to non-linear in-plane pressure.
- The findings pertaining to the impact of in-plane compressive loading type indicate that dimensionless buckling exhibits an upward trend as the load parameter χ increases, regardless of whether the loading type is linear or non-linear in distribution. Moreover, while examining the dimensionless CBL in linear and non-linear modes, it becomes evident that the nanoplate has less static stability under non-linear in-plane loading conditions.
- By increasing the hardness of the elastic substrate, the dimensionless CBL of the FGP nanoplate increases. In addition, the dimensionless CBL in the presence of the Pasternak substrate is higher than the dimensionless CBL in the presence of the Winkler foundation. The reason for this is that Winkler's substrate only considers vertical stresses, while Pasternak's medium considers transverse shear stresses in addition to vertical stresses

The next study will focus on deriving the governing equations while accounting for geometrical nonlinearities, and then analyzing the post-buckling behavior of FGP nanoplates under uniform and non-uniform in-plane loads.

CRediT authorship contribution statement

Ihab Omar: Writing – review & editing, Conceptualization, Data curation, Formal analysis, Supervision, Investigation, Writing – original draft. **Thamer Marhoon:** Writing – review & editing, Conceptualization, Data curation, Formal analysis, Supervision, Investigation, Writing – original draft. **Shahram Babadoust:** Writing – review & editing, Conceptualization, Data curation, Formal analysis, Supervision, Investigation, Writing – original draft. **Akram Shakir Najm:** Writing – review & editing, Conceptualization, Data curation, Formal analysis, Supervision, Investigation, Writing – original draft. **Mostafa Pirmoradian:** Writing – review & editing, Conceptualization, Data curation, Formal analysis, Supervision, Investigation, Writing – original draft. **Soheil Salahshour:** Writing – review & editing, Conceptualization, Data curation, Formal analysis, Supervision, Investigation, Writing – original draft. **S. Mohammad Sajadi:** Writing – review & editing, Conceptualization, Data curation, Formal analysis, Supervision, Investigation, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

[1] A. Koochi, M. Abadyan, Nonlinear Differential Equations in Micro/Nano Mechanics: Application in Micro/Nano Structures and Electromechanical Systems, Elsevier, 2020.
[2] D. Schmool, Nanotechnologies: The Physics of Nanomaterials: Volume 2: Physical Properties of Nanostructured Materials and Their Applications, Apple Academic Press, 2021.
[3] P. Phung-Van, H. Nguyen-Xuan, P.T. Hung, C.H. Thai, Nonlinear isogeometric analysis of magneto-electro-elastic porous nanoplates, Appl. Math. Model. 128 (2024) 331–346.
[4] T. Van Ke, P. Van Minh, N.T. Dung, L.M. Thai, D. Van Thom, Flexoelectric effect on bending and free vibration behaviors of piezoelectric sandwich FGP nanoplates via nonlocal strain gradient theconory, J. Vib. Eng. Technol. (2024) 1–30.
[5] A. Ghobadi, Y.T. Beni, K.K. Żur, Porosity distribution effect on stress, electric field and nonlinear vibration of functionally graded nanostructures with direct and inverse flexoelectric phenomenon, Compos. Struct. 259 (2021) 113220.
[6] T.-V. Vu, H.-L. Cao, G.-T. Truong, C.-S. Kim, Buckling analysis of the porous sandwich functionally graded plates resting on Pasternak foundations by Navier solution combined with a new refined quasi-3D hyperbolic shear deformation theory, Mech. Based Des. Struct. Mach. 51 (11) (2023) 6227–6253.
[7] M. Pirmoradian, E. Torkan, D. Toghraie, Study on size-dependent vibration and stability of DWCNTs subjected to moving nanoparticles and embedded on two-parameter foundations, Mech. Mater. 142 (2020) 103279.
[8] C. Zhang, A. Eyvazian, M. Alkhedher, M. Alwetaishi, N.A. Ahammad, Modified couple stress theory application to analyze mechanical buckling behavior of three-layer rectangular microplates with honeycomb core and piezoelectric face sheets, Compos. Struct. 292 (2022) 115582.
[9] M. Shojaeian, Y.T. Beni, H. Ataei, Electromechanical buckling of functionally graded electrostatic nanobridges using strain gradient theory, Acta Astronaut. 118 (2016) 62–71.
[10] P. Sourani, M. Hashemian, M. Pirmoradian, D. Toghraie, A comparison of the Bolotin and incremental harmonic balance methods in the dynamic stability analysis of an Euler–Bernoulli nanobeam based on the nonlocal strain gradient theory and surface effects, Mech. Mater. 145 (2020) 103403.
[11] H. Shahverdi, M.R. Barati, Vibration analysis of porous functionally graded nanoplates, Int. J. Eng. Sci. 120 (2017) 82–99.
[12] H.B. Khaniki, S. Hosseini-Hashemi, A. Nezamabadi, Buckling analysis of nonuniform nonlocal strain gradient beams using generalized differential quadrature method, Alexandria Eng. J. 57 (3) (2018) 1361–1368.
[13] V.H. Dang, T.H. Nguyen, Buckling and nonlinear vibration of functionally graded porous micro-beam resting on elastic foundation, Mech. Adv. Compos. Struct. 9 (1) (2022) 75–88.
[14] K. Khorshidi, A. Fallah, Buckling analysis of functionally graded rectangular nano-plate based on nonlocal exponential shear deformation theory, Int. J. Mech. Sci. 113 (2016) 94–104.
[15] R.A. Alghanmi, Nonlocal strain gradient theory for the bending of functionally graded porous nanoplates, Materials 15 (23) (2022) 8601.
[16] A.M. Zenkour, M.H. Aljadani, Buckling response of functionally graded porous plates due to a quasi-3D refined theory, Mathematics 10 (4) (2022) 565.
[17] M.E. Golmakani, H. Vahabi, Nonlocal buckling analysis of functionally graded annular nanoplates in an elastic medium with various boundary conditions, Microsyst. Technol. 23 (2017) 3613–3628.
[18] M.R. Barati, H. Shahverdi, Forced vibration of porous functionally graded nanoplates under uniform dynamic load using general nonlocal stress–strain gradient theory, J. Vib. Control 24 (20) (2018) 4700–4715.
[19] E. Mohseni, A. Naderi, Effect of higher order shear and normal deformations theory in buckling analysis of thick porous functionally graded plates, J. Comput. Appl. Mech. 54 (3) (2023) 347–364.
[20] C. Xu, Y. Li, M. Lu, Z. Dai, Buckling analysis of functionally graded nanobeams under non-uniform temperature using stress-driven nonlocal elasticity, Appl. Math. Mech. 43 (3) (2022) 355–370.
[21] M. Malikan, F. Tornabene, R. Dimitri, Nonlocal three-dimensional theory of elasticity for buckling behavior of functionally graded porous nanoplates using volume integrals, Mater. Res. Express 5 (9) (2018) 95006.
[22] M. Pirmoradian, E. Torkan, H. Karimpour, Parametric resonance analysis of rectangular plates subjected to moving inertial loads via IHB method, Int. J. Mech. Sci. 142 (2018) 191–215.
[23] M. Hosseini, A. Jamalpoor, A. Fath, Surface effect on the biaxial buckling and free vibration of FGM nanoplate embedded in visco-Pasternak standard linear solid-type of foundation, Meccanica 52 (6) (2017) 1381–1396.
[24] E. Torkan, M. Pirmoradian, M. Hashemian, On the parametric and external resonances of rectangular plates on an elastic foundation traversed by sequential masses, Arch. Appl. Mech. 88 (8) (Apr. 2018) 1411–1428.
[25] H. Salehipour, A.R. Shahidi, H. Nahvi, Modified nonlocal elasticity theory for functionally graded materials, Int. J. Eng. Sci. 90 (2015) 44–57.
[26] E. Torkan, M. Pirmoradian, M. Hashemian, Instability inspection of parametric vibrating rectangular Mindlin plates lying on Winkler foundations under periodic loading of moving masses, Acta Mech. Sin. 35 (1) (2019) 242–263.
[27] S. Natarajan, S. Chakraborty, M. Thangavel, S. Bordas, T. Rabczuk, Size-dependent free flexural vibration behavior of functionally graded nanoplates, Comput. Mater. Sci. 65 (2012) 74–80.
[28] P.T. Nguyen, T.V Vi, T.T. Nguyen, V.T. Vu, The plate on the nonlinear dynamic foundation under moving load, Coupled Syst. Mech. 12 (1) (2023) 83.
[29] R. Aghababaei, J.N. Reddy, Nonlocal third-order shear deformation plate theory with application to bending and vibration of plates, J. Sound Vib. 326 (1–2) (2009) 277–289.
[30] E. Torkan, M. Pirmoradian, Efficient higher-order shear deformation theories for instability analysis of plates carrying a mass moving on an elliptical path, J. Solid Mech. (2019).
[31] T.-V. Vu, A. Khosravifard, M.R. Hematiyan, T.Q. Bui, A new refined simple TSDT-based effective meshfree method for analysis of through-thickness FG plates, Appl. Math. Model. 57 (2018) 514–534.
[32] R.P. Shimpi, H.G. Patel, Free vibrations of plate using two variable refined plate theory, J. Sound Vib. 296 (4–5) (2006) 979–999.
[33] S.-E. Kim, H.-T. Thai, J. Lee, Buckling analysis of plates using the two variable refined plate theory, Thin-Walled Struct. 47 (4) (2009) 455–462.
[34] M.R. Barati, H. Shahverdi, A four-variable plate theory for thermal vibration of embedded FG nanoplates under non-uniform temperature distributions with different boundary conditions, Struct. Eng. Mech. 60 (4) (2016) 707–727.
[35] T.-V. Vu, H.-L. Cao, Deflection and natural frequency analysis of FG porous plates embedded in elastic foundations using four-variable hyperbolic quasi-3D theory, Arab. J. Sci. Eng. 48 (4) (2023) 5407–5445.
[36] T.-V. Vu, A. Khosravifard, M.R. Hematiyan, T.Q. Bui, Enhanced meshfree method with new correlation functions for functionally graded plates using a refined inverse sin shear deformation plate theory, Eur. J. Mech. 74 (2019) 160–175.
[37] T.-V. Vu, J.L. Curiel-Sosa, T.Q. Bui, A refined sin hyperbolic shear deformation theory for sandwich FG plates by enhanced meshfree with new correlation function, Int. J. Mech. Mater. Des. 15 (2019) 647–669.
[38] H.-L. Cao, T.-V. Vu, Natural frequencies analysis of functionally graded porous plates supported by Kerr-type foundations via an innovative trigonometric shear deformation theory, Int. J. Struct. Stab. Dyn. (2024) 2550236.
[39] R.M. Fenjan, R.A. Ahmed, A.A. Alasadi, N.M. Faleh, Nonlocal strain gradient thermal vibration analysis of double-coupled metal foam plate system with uniform and non-uniform porosities, Coupled Syst. Mech. 8 (3) (2019) 247–257.
[40] H.-L. Cao, T.-V. Vu, Free vibration analysis of the functionally graded porous plates with auxetic honeycomb core laid on Kerr-type elastic foundation, in: International Conference series on Geotechnics, Civil Engineering and Structures, 2024, pp. 425–433.
[41] T.-V. Vu, N.-H. Nguyen, T.-T.H. Nguyen, C.-T. Nguyen, Q.-H. Truong, U.-K. Van Tang, Free vibration analysis of FG sandwich plates on elastic foundation using a refined quasi-3D inverse sinusoidal shear deformation theory, in: ICSCEA 2019: Proceedings of the International Conference on Sustainable Civil Engineering and Architecture, 2020, pp. 1107–1117.
[42] F. Ebrahimi, M.R. Barati, Static stability analysis of smart magneto-electro-elastic heterogeneous nanoplates embedded in an elastic medium based on a four-variable refined plate theory, Smart Mater. Struct. 25 (10) (2016) 105014.
[43] J.N. Reddy, Theory and Analysis of Elastic Plates and Shells, CRC press, New York, 2006.

[44] C.W. Lim, G. Zhang, J.N. Reddy, A higher-order nonlocal elasticity and strain gradient theory and its applications in wave propagation, J. Mech. Phys. Solids 78 (2015) 298–313.
[45] M. Sobhy, Buckling and free vibration of exponentially graded sandwich plates resting on elastic foundations under various boundary conditions, Compos. Struct. 99 (2013) 76–87.
[46] S. Narendar, Buckling analysis of micro-/nano-scale plates based on two-variable refined plate theory incorporating nonlocal scale effects, Compos. Struct. 93 (12) (2011) 3093–3103.